

A localized Laguerre bootstrap toward a quadratic per-die bound

Research proof: three independent mathematical reviews completed

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Status. Three independent full mathematical reviews completed, with no gap identified; this is not formal verification. The classical Laguerre input was checked independently in the primary paper. The proof uses the reviewed comparison machinery of `../power_per_die_lower_bound.tex`. No assertion of literature novelty is made.

Theorem 1. *For every fixed $\varepsilon > 0$ there is $c_\varepsilon > 0$ such that every individual die in a tie-free permutation-fair family of n equiprobable finite dice has at least $c_\varepsilon n^{2-\varepsilon}$ faces. More generally, every monotone mixed-moment tensor with m columns has first and last row weights at most $C_\varepsilon m^{-2+\varepsilon}$.*

Throughout, assume $p_j(q) \in [0, 1]$ are nondecreasing in q , row weights $w_q > 0$ sum to one, and $\mathbb{E}_w \prod_{j \in S} p_j = 1/(|S| + 1)$ for every $S \subseteq [m]$. Use the notation of the cited proof:

$$u_j = 1 - p_j, \quad s = \sum_j u_j, \quad \ell = \lfloor m/4 \rfloor, \quad N = m - \ell, \quad \alpha = \ell/m, \quad x = \alpha s, \quad \theta = 1/16.$$

For a uniform ℓ -subset B , let $A_B = \prod_{j \in B} p_j$, $\bar{A} = \mathbb{E}_B A_B$, and $v_B = N^{-1} \sum_{j \notin B} \ell u_j$. The probability measure μ on x has row masses $(\ell + 1)w_q \bar{A}(q)$, and β has density

$$b_\ell(z) = \frac{\ell + 1}{\ell} (1 - z/\ell)^\ell \mathbf{1}_{[0, \ell]}(z).$$

The finite positive measure ν on x has row masses $m w_q$; its total mass is m . Constants may depend on explicitly fixed exponents or derivative orders, but never on m or the tensor.

Inputs already proved in the companion notes

We use the following facts, all proved in the cited power-bound note and `../rational_designs/asymptotic/weighted_success_subset_moments.tex`:

$$\max_j u_j \leq C(s + 1)/m, \quad m \Pr_w(s \leq t) \leq C(t + 1), \quad (1)$$

$$\bar{A} \leq e^{-\alpha s}, \quad \mathbb{E}_w (s + 1)^k e^{-\alpha s} \leq C_a^{k+1} k! / m. \quad (2)$$

For $s \leq m/256$, all $p_j \geq 1/2$. Under the success-weighted subset law $\Pr(B) = A_B/e_\ell(p)$, $\Delta = v_B - x$ satisfies

$$0 \leq \mathbb{E} \Delta \leq C s(s + 1)/m, \quad \mathbb{E} \Delta^2 \leq C \{s(s + 1)/m + s^2(s + 1)^2/m^2\}. \quad (3)$$

For any polynomial of degree at most N ,

$$\beta(P) = (\ell + 1) \mathbb{E}_w \mathbb{E}_B [A_B \text{Pol}_N(P)((\ell u_j)_{j \notin B})]. \quad (4)$$

If $P = Q_1 Q_2$ is a product of degree- d Laguerre sums with coefficient ℓ^1 norms B_1, B_2 , the Gaussian derivative and centered-variance bounds give the following useful tail estimate. The

part of the exact polarization Taylor series of orders $k \geq J$ has integrated absolute value at most

$$CB_1B_2(Cd/m)^{J/2}, \quad (5)$$

provided $d = o(m)$, $2d \leq N$, and $J \geq 2$ is fixed and even. Indeed the proof in the companion note gives, before summing,

$$CB_1B_2 \sum_{k=J}^{2d} (C\sqrt{d/m})^k k^{k/2} \lceil k/2 \rceil! / k!,$$

and the factorial ratio is bounded by $C^k(k+1)$. Its geometric sum proves (5); the earlier restriction $d \leq \sqrt{m}$ is unnecessary for this tail estimate.

The classical localized Laguerre estimate

For fixed $a \geq 0$, the standard normalized Laguerre functions

$$\mathcal{L}_r^a(y) = \sqrt{\frac{r!}{\Gamma(r+a+1)}} L_r^{[a]}(y) y^{a/2} e^{-y/2}$$

satisfy

$$|\mathcal{L}_r^a(y)| \leq C_a \begin{cases} (y\rho)^{a/2}, & 0 < y \leq 1/\rho, \\ (y\rho)^{-1/4}, & 1/\rho < y \leq \rho/2, \end{cases} \quad \rho = 4r + 2a + 2. \quad (L)$$

These are the standard bounds stated explicitly in Plewa [1, Section 2, immediately before (1)], with sources Muckenhoupt [2, p. 435] and Askey–Wainger [3, p. 699]. Only fixed nonnegative integer parameters a are used here. Equivalently, uniformly for $0 \leq y \leq \rho/2$,

$$|L_r^{[a]}(y)| \leq C_a e^{y/2} (r+1)^a (1 + (r+1)y)^{-a/2-1/4}. \quad (6)$$

Put

$$T_d(y) = \sum_{j < d} (1 - j/d) L_j(y) = \frac{1}{d} L_{d-1}^{[2]}(y), \quad U_d(z) = d^{-2} T_d(\theta z)^2, \quad R_d(z) = (2/(\theta d) - z) T_d(\theta z)^2.$$

For every fixed integer J , all $0 \leq k \leq J$, all sufficiently large d , and $0 \leq z \leq 2d$, (6) and $D^h T_d = (-1)^h L_{d-h-1}^{[h+2]}/d$ give

$$|U_d^{(k)}(z)| \leq C_J e^{\theta z} d^k (1 + dz)^{-k/2-5/2}, \quad (7)$$

$$|R_d^{(k)}(z)| \leq C_J e^{\theta z} d^{k+1} (1 + dz)^{-k/2-3/2}. \quad (8)$$

Here $d - h - 1$ is comparable to d , and $\theta z \leq d/8 < \rho/2$. The product rule proves (7); multiplication by $2/(\theta d) - z$ proves (8) using $2/(\theta d) + z \leq C(1 + dz)/d$. The global Laguerre coefficient products $B_1 B_2$ are $O(1)$ for U_d and $O(d)$ for R_d , by the exact identity $y T_d = d^{-1} \sum_{j < d} L_j - L_d$.

A lower-tail estimate for subset averages

Lemma 1. *On every row, for a uniform subset B ,*

$$\Pr_B(v_B < x/2) \leq \exp[-cm \min(x, 1)].$$

This statement concerns the uniform subset law. In expressions with success weighting, the unnormalized bad contribution obeys the same bound because $0 \leq A_B \leq 1$.

Proof. The whole population $z_j = \ell u_j$ has mean x and is contained in $[0, R]$ with $R = C(s+1)$ by (1). The complement is a uniform N -subset. Maclaurin's inequality and the chord bound for $e^{-z/R}$ give

$$\mathbb{E}_B e^{-Nv_B/R} \leq \left(\frac{1}{m} \sum_j e^{-z_j/R} \right)^N \leq \exp[-Nx(1 - e^{-1})/R].$$

Markov's inequality at $v_B = x/2$ now yields $\exp[-(1 - e^{-1} - 1/2)Nx/R]$. Since $N \asymp m$ and $s = x/\alpha \asymp x$, this is the claimed bound. The row $x = 0$ causes no difficulty. $\square$

Localized comparison with a distribution-function hypothesis

Proposition 1. *Fix $0 < \eta < b < 1$ and a positive constant C_0 . Suppose*

$$\nu([0, t]) \leq C_0(t + h) \quad (t \geq 0), \quad 0 < h \leq 1. \quad (9)$$

For $m^\eta \leq d \leq m^b$, a fixed even $J \geq 4$, and any fixed $A > 0$, one has, uniformly in this range,

$$|\mu(U_d) - \beta(U_d)| \leq C_J \{m^{-1}(1 + hd) + (Cd/m)^{J/2}\} + O(m^{-A}), \quad (10)$$

$$|\mu(R_d) - \beta(R_d)| \leq C_J \{(d/m)(1 + hd) + d(Cd/m)^{J/2}\} + O(m^{-A}). \quad (11)$$

The implied constants may depend on η, b, A, C_0, J .

Proof. Write $\tau = 0$ for U_d , $\tau = 1$ for R_d , and let $b_0 = 5/2$ or $3/2$, respectively. Thus the coefficient product is $O(d^\tau)$ and the fixed-order derivative bounds have the form

$$|P^{(k)}(z)| \leq C_J e^{\theta z} d^{k+\tau} (1 + dz)^{-k/2-b_0}. \quad (D)$$

We compare exact polarization to $P(v_B)$, and then $P(v_B)$ to $P(x)$.

First consider $0 \leq x \leq 1/d$. The global Gaussian-variance argument in the companion proof bounds the full polarization error pointwise by $Cd^{\tau+1}/m$, after averaging with A_B ; here $s+1 = O(1)$ and its geometric series starts at order two. Its sharp weighted Taylor argument bounds the averaging error by the same quantity, using (3); the factors singular in x are canceled by the factors s in those moment bounds. At $x = 0$ both errors vanish. Consequently the integrated contribution is at most

$$C \frac{d^{\tau+1}}{m} \nu([0, 1/d]) \leq C \frac{d^\tau}{m} (1 + hd). \quad (S)$$

Next consider $1/d < x \leq d$. This region lies in $s \leq m/256$ for large m because $d \leq m^b = o(m)$. Call a subset good if $v_B \geq x/2$. Always $v_B \leq x/(1 - \alpha) < 2x$. On good subsets the centered variance satisfies

$$\sum_{j \notin B} (z_j - v_B)^2 \leq CN(s+1)v_B.$$

Combining (D) at v_B with the centered elementary-coefficient bound, each Taylor term $2 \leq k < J$ is at most

$$C_J d^{\tau+k/2} m^{-k/2} (s+1)^{k/2} e^{\theta v_B} (1 + dx)^{-b_0}.$$

After averaging with A_B , its exponential factor is at most e^{-cx} ; the fixed power of $s+1$ can be absorbed by reducing c . Since $d/m \leq 1$, summing these fixed orders bounds the result by

$$C_J \frac{d^{\tau+1}}{m} e^{-cx} (1 + dx)^{-b_0}. \quad (12)$$

The contribution from orders $k \geq J$, integrated over all rows, is bounded separately by (5).

On bad subsets, the global Gaussian-variance bound for each of these finitely many orders is a polynomial in m times e^{Cx} . Lemma 1, with $A_B \leq 1$, bounds its average by a polynomial in m times $e^{Cx - cm \min(x, 1)}$. For $1/d < x \leq d$, this is $O(m^{-A})$ for every fixed A , uniformly: when $x \leq 1$ use $m/d \geq m^{1-b}$, and when $x \geq 1$ use $d = o(m)$. The additional integration against a measure of total mass m does not change this conclusion.

For the weighted averaging error write the exact Taylor remainder at x and retain the full mean term $P'(x)\mathbb{E}\Delta$. On good subsets the intervening points lie in $[x/2, 2x]$. Using (D) with $k = 1, 2$ and (3), then multiplying by $\bar{A}$, bounds this contribution by (12). Explicitly, before the exponential factor is absorbed, the mean term has size at most

$$C_J d^{\tau+1} (1+dx)^{-b_0-1/2} x(x+1)/m,$$

and the quadratic remainder has size at most

$$C_J d^{\tau+2} (1+dx)^{-b_0-1} \{x(x+1)/m + x^2(x+1)^2/m^2\}.$$

The former is bounded as claimed because $x/\sqrt{1+dx} \leq 1$ for $x \leq d$; for the latter use $dx/(1+dx) \leq 1$. Fixed remaining powers of $x+1$ are absorbed by the exponential margin. On bad subsets bound the remainder by $|P(v_B)| + |P(x)| + |P'(x)||\Delta|$, a polynomial in m times e^{Cx} . Multiplication by $\bar{A}$ converts its weighted probability exactly back into $\mathbb{E}_B A_B \mathbf{1}_{\text{bad}}$, bounded by the uniform bad probability. Thus Lemma 1 again gives $O(m^{-A})$.

For $x > d$ the global estimates suffice. In the integrated polarization series reserve half of the negative exponential factor to obtain e^{-cd} , and use (2) with the other half. The same geometric-series calculation gives a polynomial factor times e^{-cd} . For averaging, use the nonsingular bound $|P'| \leq C d^{\tau+1} e^{\theta z}$ and the uniform-subset Cauchy–Schwarz estimate $\mathbb{E}_B A_B |v_B - x| \leq C(s+1)e^{-\alpha s/2}/\sqrt{m}$. The exponential margin and (2) again give a polynomial factor times e^{-cd} . Since $d \geq m^\eta$, both tails are $O(m^{-A})$.

Finally, for $g(x) = e^{-cx}(1+dx)^{-b_0}$, Stieltjes integration by parts and (9) give

$$\int g d\nu \leq C_0 \int_0^\infty (t+h)(-g'(t)) dt = C_0 \left(\int_0^\infty g(t) dt + h \right) \leq C(1/d + h),$$

including a possible atom at zero. Integrate (12), combine with (S), the high-order tail and (4), to obtain the result. $\square$

The bootstrap

Lemma 2. *One has $\nu([0, t]) \leq C(t + m^{-1/2})$ for all $t \geq 0$.*

Proof. The reviewed bilinear comparison gives $\mu(U_r) \leq \beta(U_r) + Cr/m \leq C/r + Cr/m$ for $r \leq \sqrt{m}$. Here $\beta(U_r) \leq C/r$ follows by Laguerre orthogonality and $b_\ell(z) \leq 2e^{-\theta z}$. Also $U_r(z) \geq c$ for $0 \leq z \leq c_1/r$, by the elementary near-zero Laguerre estimate. Choose c_1 small enough that $x \leq c_1$ implies $s \leq 1/2$; then $\bar{A} \geq 1 - s \geq 1/2$, so ν and μ are comparable on this interval. Taking $r \asymp \min(\sqrt{m}, 1/t)$ proves the claim for $t \leq c_1$. For larger t , the original mass bound in (1) is $O(t+1) = O(t)$. $\square$

Proposition 2. *If, for a fixed $0 < a < 1$, every such tensor satisfies $\nu([0, t]) \leq C_a(t + m^{-a})$, then with $b = (1+a)/2$ one also has $\nu([0, t]) \leq C_b(t + m^{-b})$.*

Proof. Use Proposition 1 with $h = m^{-a}$ and an even J so large that $J(1-b)/2 > b$. Uniformly for degrees $m^{a/2} \leq r \leq m^b$, it gives

$$\mu(U_r) \leq C/r + C\{m^{-1} + m^{-a}r/m + (Cr/m)^{J/2}\} + O(m^{-A}) \leq C/r,$$

where $A > b$ and $r^2 \leq m^{1+a}$ were used. The same near-zero positivity and μ - ν comparison as in Lemma 2 imply $\nu([0, c_1/r]) \leq C/r$. For $t \lesssim m^{-a}$ choose $r \asymp \min(m^b, 1/t)$; these degrees are at least $m^{a/2}$ for large m . For $t \gtrsim m^{-a}$ use the old estimate. This proves the new distribution-function bound. $\square$

Starting from $a_0 = 1/2$, iteration gives $a_j = 1 - 2^{-j-1}$. Fix one such a , set $b = (1 + a)/2$, and now choose $d = \lfloor \kappa m^b \rfloor$ in (11), with $\kappa > 0$ sufficiently small. Choose J as in Proposition 2. The high-order tail is $o(1)$ and

$$(d/m)(1 + m^{-a}d) = o(1) + O(\kappa^2).$$

The scalar calculation from the reviewed power-bound proof gives $\beta(R_d) \geq 1/(96\theta) > 0$. Hence $\mu(R_d) > 0$ for sufficiently small κ and large m . Some row has $x < 2/(\theta d)$, and the last row minimizes x , so $x_K < 2/(\theta d)$. The already proved distribution-function estimate with exponent b then gives

$$mw_K \leq \nu([0, 2/(\theta d)]) \leq C_b/d, \quad w_K \leq C_b m^{-1-b}.$$

Reverse the rows and replace every p_j by $1 - p_j$ for the first row. Since the exponents b approach one, this proves Theorem 1. The constants are not uniform as $\varepsilon \downarrow 0$; a quadratic $\Omega(n^2)$ lower bound is not asserted.

References

- [1] P. Plewa, Hardy's inequality for Laguerre expansions of Hermite type, *J. Fourier Anal. Appl.* 25 (2019), 1855–1873. <https://doi.org/10.1007/s00041-018-9642-2>.
- [2] B. Muckenhoupt, Mean convergence of Hermite and Laguerre series. II, *Trans. Amer. Math. Soc.* 147 (1970), 433–460.
- [3] R. Askey and S. Wainger, Mean convergence of expansions in Laguerre and Hermite series, *Amer. J. Math.* 87 (1965), 695–708.